

Sensors

There are a multitude of sensors in smart phones these days which has made the device even more versatile. There are light sensors, barometers, hygrometers, compass to name a few. Let's have a detailed look at this.

ALS

An ambient light sensor makes use of two different types of photodiodes (or light sensors) to each record the level of light currently falling on the screen. By combining the output of these two sensors portable devices such as tablets, smartphones, and laptops can sense a side pectrum of light. Thus it can accurately measure the brightness of the room. This information is then used to adjust the screen brightness to make it easier on the user's eyes and also helps to extend battery life.

Proximity Sensor

A proximity sensor detects how close a smartphone is to your body thus letting the smartphone know whether you have the screen close to your cheek. In order to save battery life and also to prevent accidental touchscreen input the phone automatically switches these things off. Thus the user can take a call by just placing the phone on their cheek, there is no need to do anything else. All current tasks will be saved and the display switched off for the duration of the call.

Accelerometer

An accelerometer is what reads the acceleration of the device along three axes - X, Y and Z. So when you tilt your phone or flick your phone to change a radio station or skip a song, it is the accelerometer which senses this and performs the appropriate action. These are very sensitive devices which can measure acceleration up to 1G on a fine scale.

Compass

The compass is simply used to figure out the magnetic north and relay the information. However, most electronic circuits these days do not use magnets but instead they rely on the change in a magnetic field around a flow of current to figure out the direction. This is why your compass application demands a calibration each time it is activated. Without the compass sensor your phone will not be able to provide you with directions even if you have the application installed.